



Housekeeping

- Workshop 3

Help seeking?

- [EdStem Discussion](#)

Computing Environments: Today's Learning Objectives

- What is “Bit rot”?
- What is hard about managing computing environment?
- Learn about different types of computing environments
- Basic intuition for how VMs work

Computing Environments

& Virtualization Concepts

Does code rot?

Unfathomed misunderstanding is further revealed by the term “software maintenance”, as a result of which many people continue to believe that programs—and even programming languages themselves—are subject to wear and tear. Your car needs maintenance too, doesn’t it? Famous is the story of the oil company that believed that its PASCAL programs did not last as long as its FORTRAN programs “because PASCAL was not maintained”.

— Edsger W. Dijkstra, [On the cruelty of really teaching computing science](#)

*Before Borland’s new spreadsheet for Windows shipped, Philippe Kahn, the colorful founder of Borland, was quoted a lot in the press bragging about how Quattro Pro would be much better than Microsoft Excel, because it was written from scratch. All new source code! As if source code **rusted**.*

*The idea that new code is better than old is patently absurd. Old code has been used. It has been tested. Lots of bugs have been found, and they’ve been fixed. There’s nothing wrong with it. It doesn’t acquire bugs just by sitting around on your hard drive. Au contraire, baby! Is software supposed to be like an old Dodge Dart, that rusts just sitting in the garage? Is software like a teddy bear that’s kind of gross if it’s not made out of **all new material**?*

— Joel Spolsky, [Things You Should Never Do, Part I](#)





Whois Record for FaceBook.com

Domain Available

facebook.com is for sale!

This domain is listed for sale on our partner sites .

Visit our partner to buy facebook.com

Domain Profile

Registrant REDACTED

decade rollover bugs



Investigating issues with Listing Edit
@etsystatus.com

2:05 PM · Jan 2, 2020 · Twitter Web App



Kara Swisher @karaswisher · Oct 4, 2021
Understanding How Facebook Disappeared from the Internet

CLOUDFLARE BLOG

What Dijkstra and Spolsky are missing is that *the world changes around software*. Software doesn't exist in a vacuum: it's part of an ecosystem. Legacy systems have legacy dependencies, and run in legacy environments. Those dependencies and environments are not static, they change over time, and sometimes the old ones go away, or are too expensive or risky to keep using.

Software is indeed different from physical artifacts, in that software artifacts (source code, binaries) don't change with use. But in the world of software, that's exactly the problem. The world keeps changing, and the software doesn't, unless you put the work into it. And, unlike civil engineers, we aren't yet good at thinking about the intended lifetime of a software system when we're designing it.

today: github.com/StackExchange/...

It's a narrow set of conditions to trigger, but ultimately we beat the ever living crap out of the thread pool after certain network events interrupted Redis commands.

StackExchange/StackExchange.Redis



#1847 Message Backlog:
Task pool exhaustion
scenario with...

ments

Craver opened on August 24, 2021



a poor design decision I made on Bo
own @Roblox for 3 full days last year
y to all the folks at Roblox and
that probably pulled all-nighters to de



Jef Poskanzer @jef_poskanzer · Oct 4, 2021
Apparently a non-trivial
common Y2K "fix" just postponed things by 20 years, interpreting 00-99 as
the 2000s and 20-99 as the 1900s. 20 years is now up, some of those
systems are still in use, and they think it's 1920. Oooops.

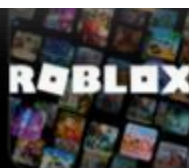
195

4.3K

11.2K



Show this thread



blog.roblox.com

Roblox Return to Service 10/28-10/31 2021 - Roblox Blo
Starting October 28th and fully resolving on October 31
Roblox experienced a 73-hour outage. We're sharing the

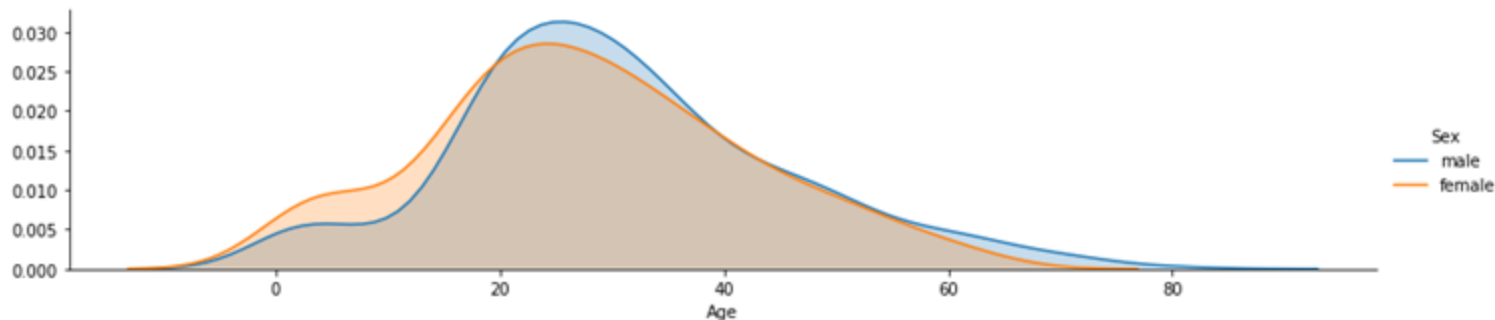
5:15 PM · Jan 20, 2022 · Twitter Web App

Even simple code snippets can be difficult to run...

```
In [ ]: titanic_df=pd.read_csv('input/train.csv')
```

```
In [ ]: fig=sns.FacetGrid(titanic_df,hue='Sex',aspect=4)
fig.map(sns.kdeplot,'Age',shade=True)
oldest=titanic_df['Age'].max()
xlim=[0,oldest]
fig.add_legend()
```

```
Out[16]: <seaborn.axisgrid.FacetGrid at 0x19f662f9f60>
```

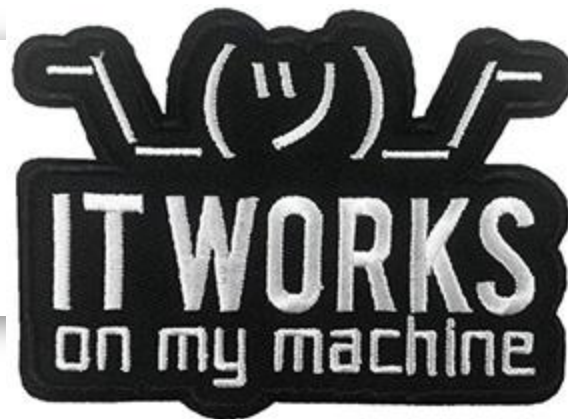


Often, the default experience is failure.

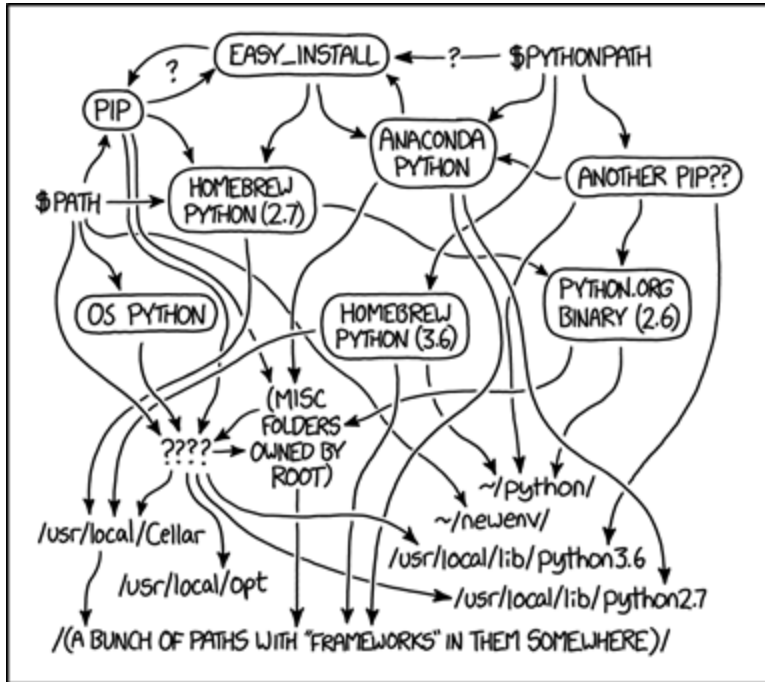


```
ModuleNotFoundError                                Traceback (most recent call last)
<ipython-input-2-3a2bf2a4b8fb> in <module>
      3 import pandas as pd
      4 import matplotlib.pyplot as plt
----> 5 import seaborn as sns
      6 get_ipython().run_line_magic('matplotlib', 'inline')

ModuleNotFoundError: No module named 'seaborn'
```



In general, managing environments is hard.



MY PYTHON ENVIRONMENT HAS BECOME SO DEGRADED
THAT MY LAPTOP HAS BEEN DECLARED A SUPERFUND SITE.



Jean Yang
@jeanqasaur

Following

Hm, that the version situation has gotten so bad that even when I'm not coding in Python, but simply using stuff packaged with scripts written in Python, I am spending nontrivial time sys-admining just to do basic things.

Programmers must be “end-user” system admins

Kenneth Reitz @kennethreitz

What are the biggest problems, as a whole, that Python is facing today?

1:21 AM - 11 Mar 2019

5 Retweets 47 Likes



Many replies with “dependencies”



Anavari Tayal @ayalavari
Replying to @kennethreitz

Lack of standardization for dependency management. We've got venv, pipenv, conda env, etc. Node has this much easier with node_modules.

1 1 31



Thomas @__tosh · Mar 11

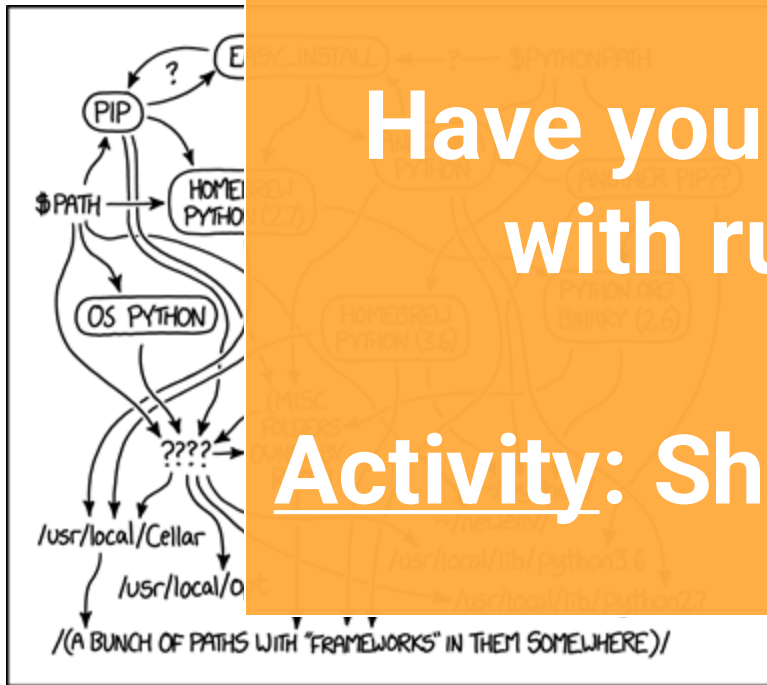
Replying to @kennethreitz @ixek

dependency/package mgmt, slow, GIL, v2/3 split

In general, managing environments is hard.

Have you ever had issues
with running code?

Activity: Share with a partner.



MY PYTHON ENVIRONMENT HAS BECOME SO DEGRADED
THAT MY LAPTOP HAS BEEN DECLARED A SUPERFUND SITE.



Anavari Tayal @ayalasanavari
Replying to @kennethreitz

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1 1 31



Thomas @__tosh · Mar 11

Replying to @kennethreitz @ixek

dependency/package mgmt, slow, GIL, v2/3 split

d-user"

many replies with dependencies



Your billion dollar software idea... is worth...

\$0.

Software on your machine has no *real* value.



Nader Dabit @dabit3 · 9h



Recruiter: do you have a link to some projects that you've built?

Me: sure - localhost:3000, localhost:3001, localhost:3030, localhost:8080



76



656



5.9K



Where does your code live and run?

Need to consider not only *code*, but it's

- Dependencies
- Environment
- Software Ecosystem.
- Stakeholders and Users
- Value

A more holistic view: Rather than thinking of code alone, think of building **computing environments**, a unified view of code and configuration needed to operate and maintain it's value.



```
FROM python:2.7.13
VOLUME /output
ENV MPLBACKEND Agg
RUN apt-get update
RUN apt-get install -y graphviz
RUN pip install pygraphviz
RUN pip install matplotlib
RUN pip install networkx
ADD draw_graph.py /snippets/
CMD python /snippets/draw_graph.py
```

```
# Import modules from networkx and matplotlib
import matplotlib.pyplot as plot
import networkx as nx
from networkx.drawing.nx_agraph import
    graphviz_layout

# Generate the complete graph on five vertices
k5 = nx.complete_graph(5)

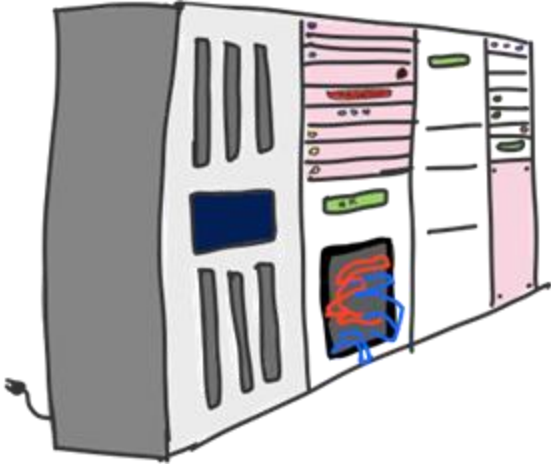
# Draw using layout generated by graphviz
plot.figure()
nx.draw(k5, graphviz_layout(k5, prog="neato"))
plot.savefig('/output/graph.png')
plot.close()
```

Configuration + Code

Resilient and scalable software co-exists with
infrastructure, i.e. computing environments.

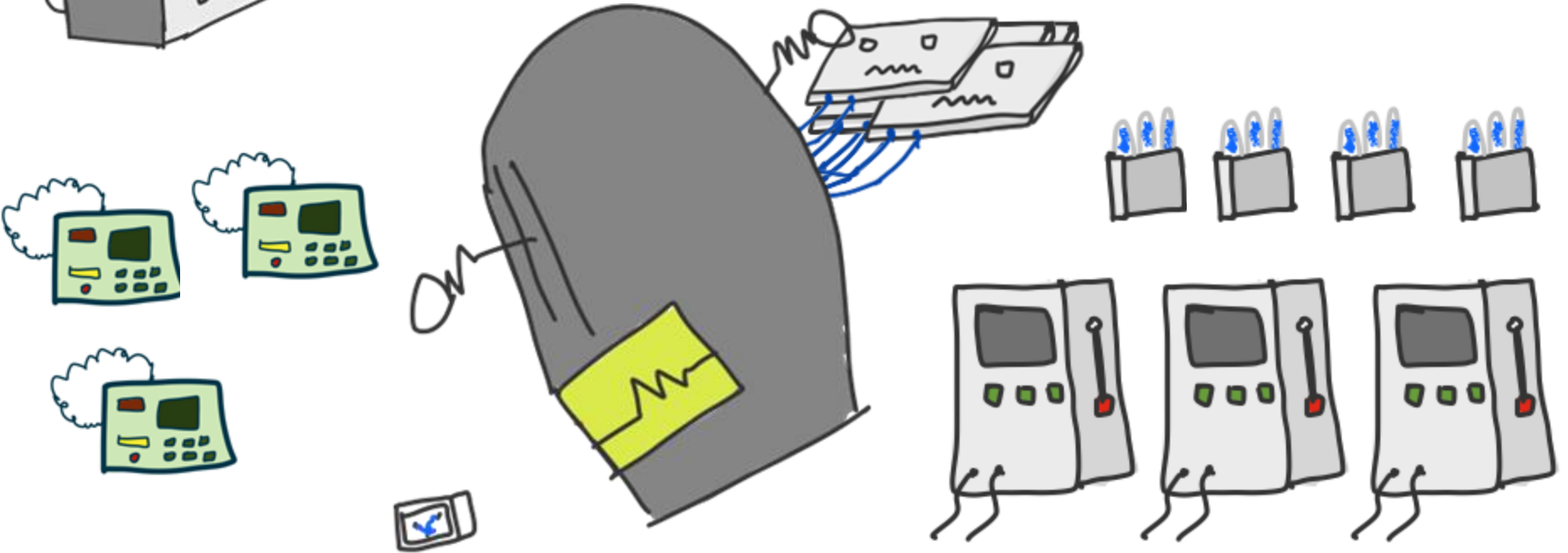
What are the properties of an ideal computing environment? (Fill-in-the-blank)

- Reproducible config and code can work in another context
- Resilient (errors, etc.)
- Secure security protocols, security secrets
- Scalable resource provisioning should be dynamic
- Flexibility How tightly coupled is your code to a cloud environment
- Maintain quality of service define quality measurement - how fast is the API?
- Resource utilization



There are countless kinds of computing environments.

You will learn how to design, configure, and operate them.



Virtualization Concepts

Essential computing Environment building block.

Virtualization is not containerization

Classic Virtual Machines.

Takes a lot of compute - Hardware communication

Fully loaded
operating systems.

Graphical
interfaces.

Optimized.

Heavy-weight.



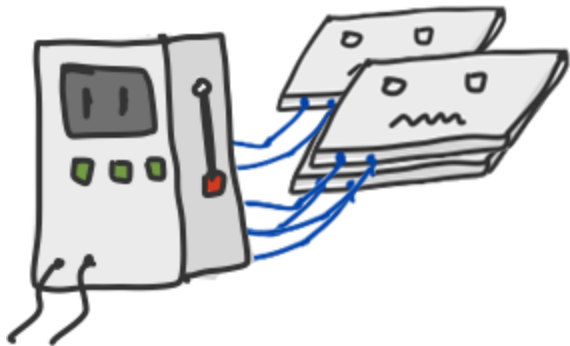
Headless Infrastructure

“headless” like *headless browsers* for integration testing.



headless VMs,
microvms, and
containers...

No Graphical Interface



You can build, use, and throw-away.



And like the 2GB VM running in the cloud...

Terms

Popek and Goldberg (1974). "A virtual machine is taken to be an efficient, isolated duplicate of the real machine".

Host/Hypervisor: Operating system running on the physical machine.

Guest: Operating system running in the virtual machine.

Virtual machine monitor (VMM): Helps run guest on host. VMM implements the virtualization interface, enforces the illusion of isolated virtual machines

Examples of Privileged Instructions include:

- a) Perform I/O operations
- b) Memory address mapping
- c) Load and read system registers
- d) Change processor modes from kernel to user

Full emulation

Hosted interpretation

```
while(1){
curr_instr = fetch(virtHw.PC);
virtHw.PC += 4;
switch(curr_instr){
case ADD:
    int sum = virtHw.regs[curr_instr.R0] +
virtHw.regs[curr_instr.R1];
    virtHw.regs[curr_instr.R0] = sum;
    break;
case SUB:

    //...etc...

}
}
```

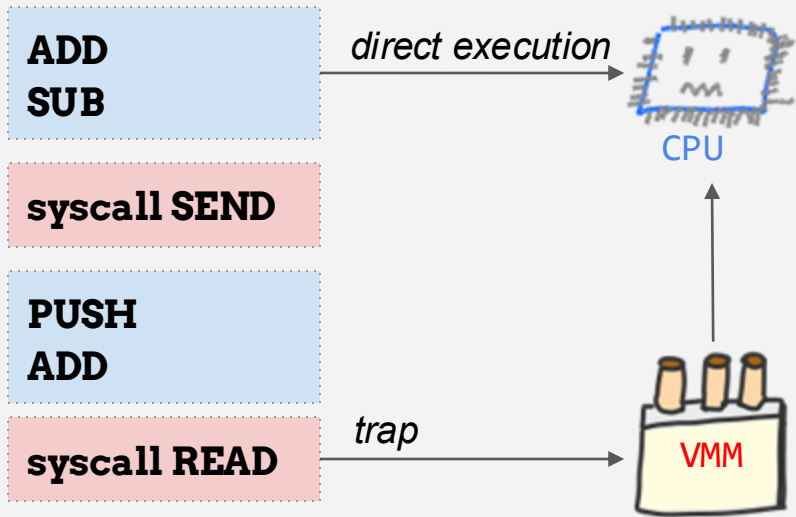
Run virtual machine as a User Application.

- *Emulate CPU*
- *Emulate Memory, Disks*
- *Emulate Network, other hardware...*

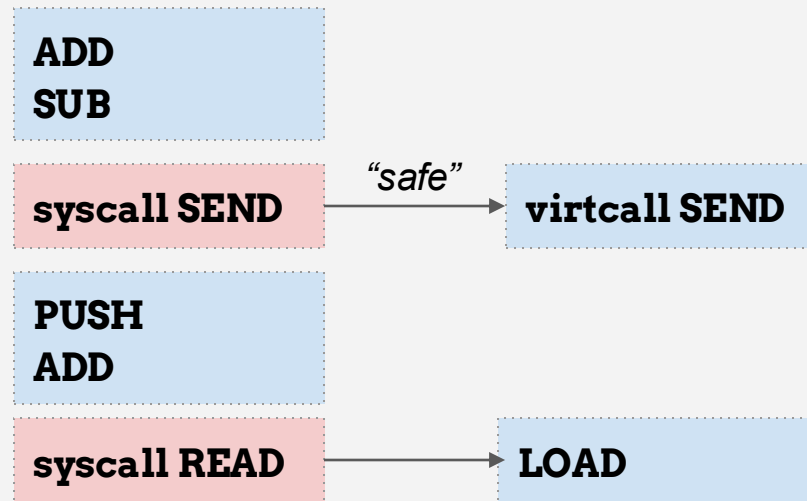
Problems?

Optimizations.

Trap-and-emulate



Binary translation



Hardware-assisted virtualization



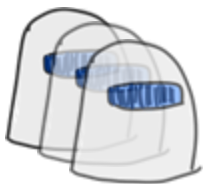
User mode

**ADD
SUB**



Kernel mode

syscall SEND



Guest mode

VMREAD



VT-x CPU

 *New instructions allow virtual machine operations to be more efficiently supported.*

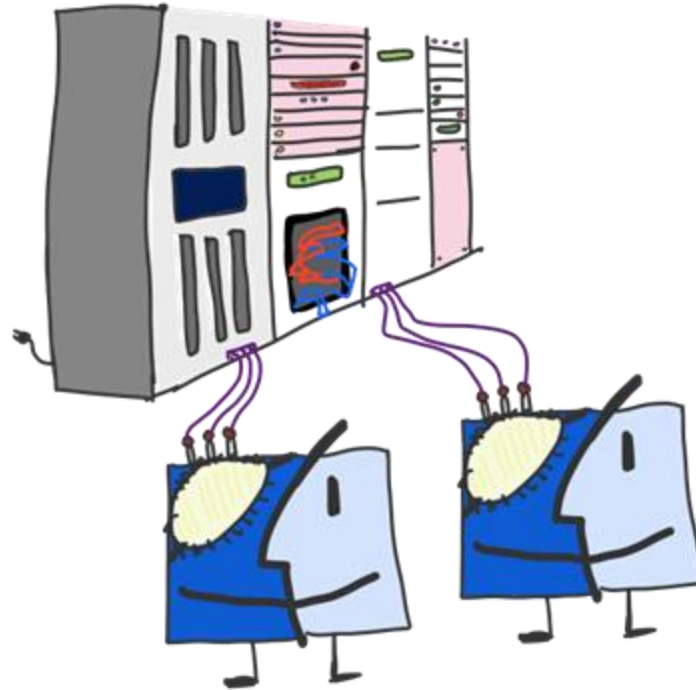
 *VMM is also easier to implement.*

Para-virtualization



Patch Guest OS in order to enable virtualization via VMM.

VMM



Examples

Full emulation

[Bochs, PDP-10, Video games]

Hardware-assisted

[KVM, ESXi, Hyper-V]

Trap-and-emulate

[System/370]

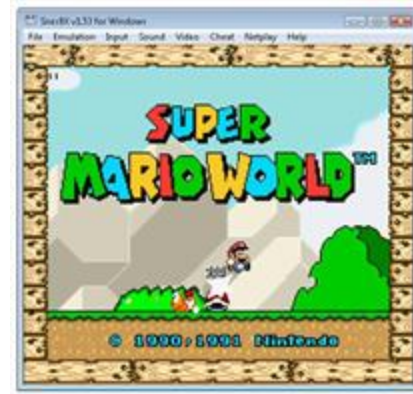
Paravirtualization

[Xen, Hyper-V, VF, Hyperkit]

Binary translation

[Qemu, VMWare, VirtualBox]

Note: Apple “silicon” M-class laptops use ARM-based processors, which do not support hardware visualization...



Trends

AWS moved from **Xen** (paravirtualization) => **KVM** (hardware assisted virtualization).

<https://medium.com/@dbclin/aws-just-announced-a-move-from-xen-towards-kvm-so-what-is-kvm-2091f123991>

OS-level Virtualization

The operating system provides mechanisms for virtualizing resources.

You are not performing full virtualization of a system.

JULIA EVANS
@b0rk

chroot

a container image is a
tarball of a filesystem
(or several tarballs - 1 per layer)

if someone sends me
a tarball of their
filesystem how do
I use that though?

chroot: trick a program into thinking
it has a different root directory

```
$ ls /path/to/container_filesystem
bin/ etc/ usr/ var/
$ sudo chroot /path/to/container_filesystem /bin/bash
(inside chroot now)
$ ls /
bin/ etc/ usr/ var/ ← that's our new fake root directory!
                        we tricked ls!
```

very basic way to
run a Redis container
tarball of filesystem
with Redis installed

```
$ mkdir redis; cd/redis
$ tar -xzf redis.tar
$ chroot $PWD /usr/bin/redis
# done! redis is running!
```

problems with just
using chroot

- no CPU/memory limits
- other running processes are still visible
- can't use the same network port as another process
- LOTS of security issues

Docker uses pivot-root
+ extra isolation features
to run containers

pivot-root is like chroot
but harder to escape from

JULIA EVANS
@b0rk

processes can use
a lot of memory

process: I want 10 GB of memory
process: me too!
guys I only have 16 GB total! Linux

use too much memory:
get OOM killed

process: I want 1 GB of memory
Linux: NOPE your limit was 500 MB you die now
process: oh no

cgroups

a cgroup is a
group of processes

cgroup!
Usually you'll assign
the same cgroup
to every process
in a container.

cgroups have
memory/CPU limits

you three get 500 MB
of RAM to share, okay?

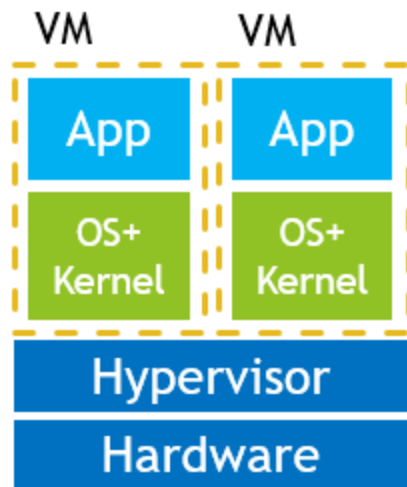
use too much CPU:
get slowed down

process: I want to use ALL THE CPU!
you hit your quota this
millisecond, you'll have
to wait Linux

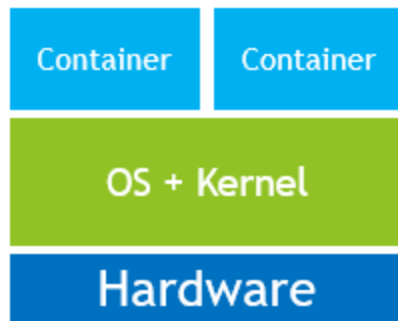
cgroups track
memory & CPU usage

that cgroup is
using 412.3 MB of
memory right now!
see /sys/fs/cgroup!

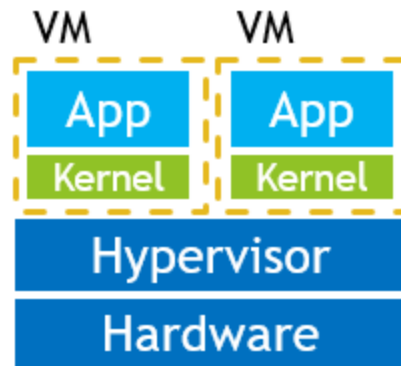
VMs vs. Containers vs. Unikernels



Virtual Machines



Linux Containers



Unikernels

No one best computing environment

Baremetal

Use case:

Desktop/Mobile/Embedded computing.

Efficiency/CI Harness

Headless Virtualization

Use case:

General purpose computing services.

Micro VMs

Use case:

Embedded appliances, such as Docker for Mac, Windows Subsystem Linux (WSL)

Os-level Virtualization

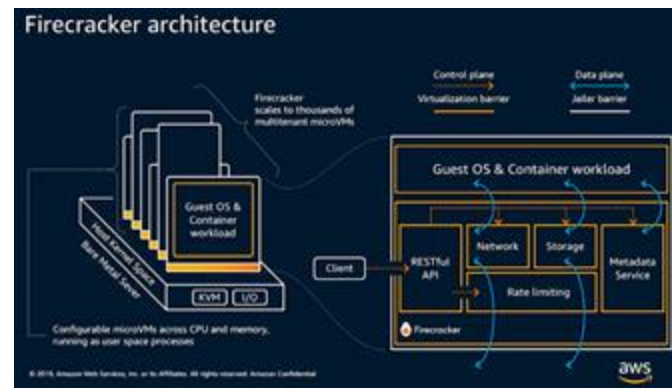
Use case:

Self-contained processes.

Unikernels

Use case:

Lambda/Optimized ML FPGAs (Azure)



2025: [Firecracker is still a foundation of AWS](#)

Additional Resources

- “[Automated Software Engineering](#)” conference
- [DevOpsDays](#)